Credit Risk and Loan Performance

EXECUTIVE SUMMARY

This report presents a comprehensive evaluation of retail loan default risk, borrower credit profiles, and portfolio performance metrics based on an analysis is to identity key risk drivers, evaluate current underwriting and risk-based pricing efficiency, and outline strategic interventions to mitigate default exposure while optimizing loan yield.

Key Metrics Overview

- Total Portfolio Records Analyzed: **32,581** loans.
- Historical Default on File Rate: **17.63%** of borrowers (**5,745** out of **32,581**)
- Default Rate for Historical Defaulters: **37.81%** (vs. **18.39%** for clean credit files)
- Critical Debt Burden Tipping Point: **30%** Loan-to-Income Ratio (default rate surges from **21.95%** to **70.32%**)
- Highest-Risk Loan Intent: Debt Consolidation (**28.59%** default rate)
- Highest-Risk Homeownership Segment: Renters (**31.57%** default rate)

PROJECT OVERVIEW

The primary dataset
(**Credit_risk-dataset.csv**) contains granular borrower demographic details, loan parameters, credit bureau indicators,and historical repayment records.

Core Feature Breakdown

1. **Borrower Demographics & Financial Capacity:**
   - person_age: Borrower age.
   - person_income: Annual verified gross income.
   - person_home_ownership: Residential status (**RENT, MORTGAGE, OWN, OTHER**).
   - person_emp_length: Length of continuous employment in years.
2. **Loan Structure & Price:**
   - loan_intent: Stated purpose of the loan (**DEBTCONSOLIDATION, MEDICAL, HOMEIMPROVEMENT, PERSONAL, EDUCATION, VENTURE**).
   - loan_Grade: Internal credit risk tier from **A** (**lowest risk**) to **G** (**highest risk**).
   - loan_amnt: Total principal extended.
   - loan_int_rate: Contractual interest rate percentage.
   - loan_percent_income: Debt burden ratio relative to annual income.
3. **Credit History & Target Variable:**
   - cb_person_default_on_file: Flag indicating prior default history on bureau records (**Y/N**).
   - cb_person_cred_hist_length: Years of active credit bureau history.
   - loan_status: Target binary output (**0 = Non-Default, 1 = Default**).

KEY INSIGHTS & DETAILED FINDINGS

Prior Bureau Defaults

(cb_person-default_on_file) as a Risk Indicator
Prior credit history is one of the strongest single predictors of future loan default in this portfolio.

	cb_person_default_on_file	total_borrowers	avg_interest_rate	default_rate_pct
0	N	26836	10.258913	18.393203
1	Y	5745	14.513800	37.806789

- **Insight:** Borrowers with a prior default on record default at more than double the rate of clean borrowers (37.81% vs. 18.39%). While the average interest rate is elevated by ~ 425 bps for prior defaulters (14.51% vs 10.26%), it remains insufficient to offset the 105% relative jump in default frequency.

Impact of Loan Burden Ratio

(loan_percent_income) Analyzing loan amount relative to income reveals a steep, non-linear risk curve.

	loan_income_tier	total_loans	default_rate_pct
0	< 10%	10475	11.732697
1	10% - 20%	12066	15.108570
2	20% - 30	6197	21.946103
3	> 30%	3834	70.318206

- **Insight:** The **30%** loan-to-income threshold represents an acute risk boundary. When the debt burden exceeds **30%** of annual income, default probability skyrockets by **over 2205** compared to the **20-30%** tier, reaching an unsustainable **70.32%** default rate.

Loan Intent Risk Segmentation

(loan_intent)
Borrowing motives significicantly influence repayment discipline.

	loan_intent	total_loans	total_loan_amount	avg_interest_rate	default_rate_pct
0	DEBTCONSOLIDATION	5212	50008550	10.983268	28.587874
3	MEDICAL	6071	56214925	11.060194	26.700708
2	HOMEIMPROVEMENT	3605	37349675	11.201152	26.102635
4	PERSONAL	5521	52856800	10.998221	19.887702
1	EDUCATION	6453	61191725	10.950261	17.216798
5	VENTURE	5719	54809625	10.948275	14.810282

- **Insight:** Debt Consolidation and Medical loans pose the highest portfolio risk with default rates of 28.59% and 26.70%, respectively. Conversely, Venture and Education loans show superior credit performance with default rates under 18%.

Credit Rating Grade Calibration

(loan_Grade)
Evaluation of risk tiers demonstrates clear separation in risk identification, but highlights critical pricing imbalances in extreme grades.

	loan_grade	total_borrowers	total_loan_amount	avg_interest_rate	default_rate_pct
0	A	10777	92027750	7.327651	9.956389
1	B	10451	104462800	10.995555	16.275954
2	C	6458	59503125	13.463542	20.733973
3	D	3626	39339350	15.361448	59.045780
4	E	964	12450875	17.009455	64.419087
5	F	241	3546875	18.609159	70.539419
6	G	64	1100525	20.251525	98.437500

- **Insight:** Risk progression from **Grade A** through **C** is smooth. However, starting at **Grade D (59.05%)**, defaults breakout severely. **Grade G** displays a catastrophic **98.44%** default rate. The risk premium charged **(20.25%** interest**)** is vastly inadequate to absorb loss rates exceeding **90%**.

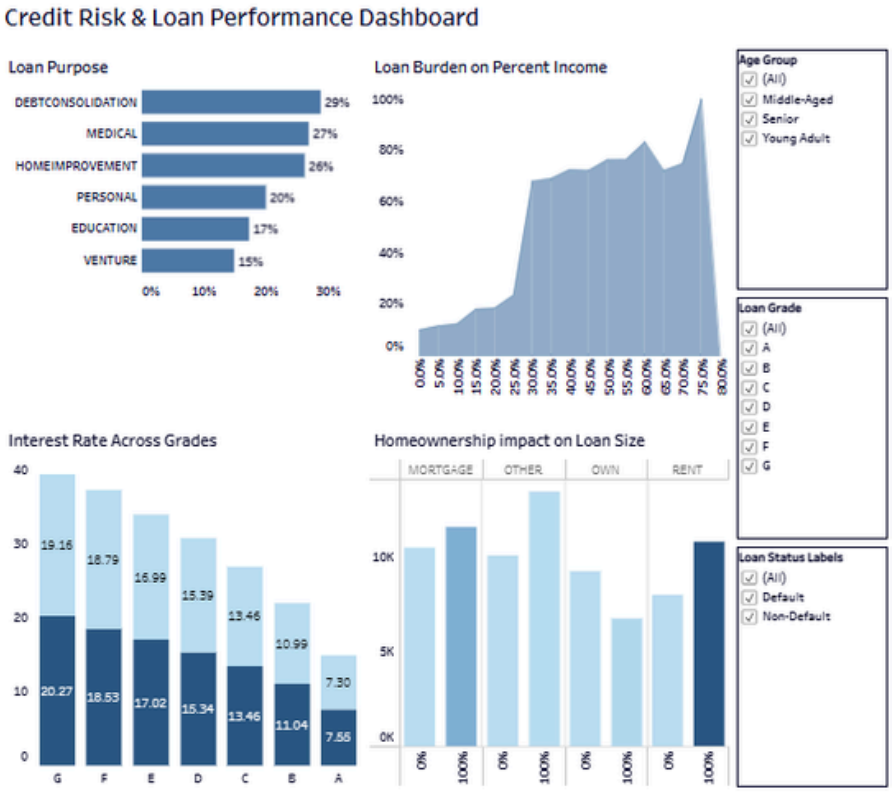
Homeownership & Financial Stability

(person_home_ownership)
Residential stability acts as a strong proxy for overall credit risk.

	person_home_ownership	total_borrowers	avg_income	avg_loan_amount	default_rate_pct
3	RENT	16446	54997.747963	8862.331266	31.569987
1	OTHER	107	76387.803738	11074.532710	30.841121
0	MORTGAGE	13444	81127.121690	10574.460726	12.570663
2	OWN	2584	57834.812693	9029.943885	7.469040

- **Insight:** Borrowers who own their homes outright exhibits the lowest default rate at 7.47%, followed closely by those with a mortgage (12.57%). Renters present a 31.57% default rate, driven by lower average annual incomes ($54,998) and higher relative financial fragility.

BUSINESS DASHBOERD



BUSINESS RECOMMENDATIONS

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